

Mini-4

Instructions. This mini-assignment contains two problems. **Choose one.** No implementation is required. Each answer should be a short, precise argument such as a paragraph or a worked calculation. We are testing whether you can reason about *why* an algorithm breaks, not whether you can run it.

Problem 1: Belief Tracking and QMDP

A rescue drone flies over a 5×5 grid. A survivor's location s^* is fixed and unknown. The drone knows its own position exactly and maintains a belief $b_t \in \Delta(\mathcal{S})$. At each step it receives a binary ping: $\Pr(o=1 \mid \text{at } s^*) = p_{\text{hit}} = 0.9$, $\Pr(o=1 \mid \text{elsewhere}) = p_{\text{fp}} = 0.1$.

Questions

1. Belief update by hand.

The drone starts with a uniform belief over all 25 cells, moves to cell (3,3), and receives $o_1 = 1$.

- (a) Write the Bayesian update $b_1 = \tau(b_0, o_1=1, x_1=(3,3))$ in closed form and compute $b_1((3,3))$ versus $b_1(s)$ for any $s \neq (3,3)$. Show your work; you may leave the answer as a fraction.
- (b) The drone stays at (3,3) and receives ten more readings, all $o = 0$. Without computing each step explicitly, describe what happens to $b((3,3))$ and explain the asymmetry between how quickly a ping concentrates belief versus how quickly repeated misses disperse it. What parameter controls that asymmetry, and in which direction?

2. QMDP's blind spot.

QMDP selects actions by $a_t = \arg \max_a \sum_s b_t(s) Q^*(s, a)$, where Q^* is the optimal value function of the *fully observable* shortest-path MDP.

- (a) The drone is at (1,1) and b_t is uniform over exactly two cells: (2,2) and (4,4). Argue qualitatively, without computing Q^* exactly, what action QMDP takes and why a belief-aware agent would prefer a different first move.
- (b) Suppose the prof proposes for you to try diagnosing QMDP failures by logging $H(b_t)$ (belief entropy) alongside the action taken each step. Describe the *specific pattern* in those two logged quantities that would indicate QMDP is failing in the way you identified above, and explain why that pattern is a reliable signal of the failure mode rather than just normal operation.

Problem 2: Credit Assignment in Cooperative MARL

Two agents navigate a 6×6 grid. The team receives $r_t = +10$ only when *both* agents simultaneously occupy their respective target cells; otherwise $r_t = 0$. Each agent observes only a 3×3 window around itself and acts independently at execution time.

Questions

1. Diagnosing credit assignment failure.

- (a) Agent 1 reliably reaches its target by step 10 of every episode, yet the team rarely earns the reward because Agent 2 is slow. Explain why, under a shared reward and policy-gradient updates, Agent 1's policy may *degrade* over training even though its individual behavior is correct.
- (b) Describe one metric, beyond team return, that you would log during rollouts to distinguish this failure mode (good individual behavior, poor coordination) from the alternative failure mode where neither agent has learned to navigate at all. Be specific about what you measure, at what point in the episode, and what each failure mode looks like in that metric.

2. QMIX's monotonicity constraint.

QMIX enforces $\frac{\partial Q_{\text{tot}}}{\partial Q_i} \geq 0$. Informally, the global greedy action must also be locally greedy for each agent.

- (a) Fill in a 2×2 joint-action value table (integer entries) where the optimal joint action is *not* a Nash equilibrium under individual greedy play, and therefore cannot be represented by QMIX. No derivation needed: just exhibit the table and identify the problematic entry.
- (b) Does the monotonicity constraint actually hurt QMIX on the two-target task described above, or is that task within the representable class? Justify your answer in two to three sentences by checking whether the optimal joint policy satisfies the IGM condition.